

Programme	B. Sc. Computer Science				
Course Title	Software Project Management				
Type of Course	Major				
Semester	III				
Academic Level	200 - 299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. Computer Science knowledge 2. Understanding fundamental computer science concepts, data structures, and algorithms 3. Basic knowledge of project planning and scheduling				
Course Summary	Students are introduced to the concepts, procedures, and resources of software project management in this course. Project scheduling, budgeting, quality assurance, risk management, and teamwork are among the subjects covered. The goal of the course is to equip students with the skills necessary for efficient project management in software development settings.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Define and explain the fundamental concepts, principles, and terminologies related to software project management. Differentiate between various software engineering process models. Understand the agile principle and methodologies and appreciate the need for iterative approaches to software Development	U	C	Instructor-created exams / Quiz

CO2	Master various design concepts used during project development life cycle.	U	P	Assignments/ Test papers/ Viva Voce
CO3	Master various SPM techniques	U	P	Seminar Presentation / Group Tutorial Work/ Viva Voce
CO4	Develop project plans, Create project schedules using tools like Gantt charts and network diagrams	Ap	C	Instructor- created exams / Home Assignments
CO5	Understand the importance of quality in software development by mastering quality assurance processes, methodologies, and testing strategies.	U	P	Writing assignments/ Exams
CO6	Prepare and deliver effective project presentations.	Ap	P	Case Study/ mini Project/ Seminar Presentation/ Group Presentations
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48+12)	Marks (70)
I	Introduction to Software Engineering and Process Models		10	12
	1	Software and Software Engineering- nature of software, Software Engineering, Software Process	2	

	2	Software Development Life Cycle (SDLC)	2	
	3	Prescriptive Process Model- Water fall model, Incremental Model, Evolutionary Process Model	2	
	4	Agile Development- What is Agility, What is agile Process?	2	
	5	Extreme Programming	2	
II		Software requirements and Design Concepts	16	22
	6	Understanding requirements- requirement engineering process	3	
	7	Feasibility studies	1	
	8	Design Concepts- Design process, Design Concepts	2	
	9	Design Model Elements- Data design elements, Architectural design elements, Interface Design Elements, Component-Level Design Elements, Deployment-Level Design Elements	2	
	10	Architectural design using DFD	2	
	11	Component level design guidelines	2	
	12	Modelling with UML – Class diagram Use Case Diagram, State chart Diagram, Activity Diagram,	4	
III		Software Project Management	11	18
	13	Introduction to Software Project Management- Overview of software project management, Importance of project management in software engineering, Role of a project manager	2	
	14	Project Planning and Scope Management- Work breakdown structure (WBS) and project estimation techniques	2	
	15	Project Scheduling and Resource Allocation- Gantt charts and network diagrams,	2	
	16	Critical Path Method (CPM) and Program Evaluation and Review Technique (PERT)	2	
	17	Risk Management-reactive vs proactive risk strategies, Risk identification, risk projection, RMMMM plan	3	
IV		Software Quality Assurance	11	18
	18	Quality Concepts- Software quality, Achieving Software quality,	2	
	19	Testing Strategies	2	
	20	Software testing- levels of software testing	1	

	21	Types of software test- Unit testing, Integration testing, Black box testing, white box testing, System testing	4	
	22	Art of debugging	2	
V	Open Ended Module- Trends in Software Engineering		12	
	1	• Case study of CASE tools • Prepare a project report • Analysis of real-world software project management case studies • Group project presentations		

References

- Roger S, “Software Engineering – A Practitioner’s Approach”, seventh edition, Pressman, 2010.
- Pearson Education, “Software Engineering by Ian Sommerville”, 9th edition, 2010.
- Pankaj Jalote, An Integrated Approach to Software Engineering, 3rd Edition, Narosa Publishing House.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	1	1	-	-	3	-						
CO 2	1	1	2	-	3	-						
CO 3	1	1	-	-	3	-						
CO 4	1	1	-	-	3	-						
CO 5	1	1	-	-	3	-						
CO 6	1	1	-	-	3	-						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)

- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	End Semester Examinations
CO 1	✓		✓
CO 2	✓		✓
CO 3	✓		✓
CO 4		✓	✓
CO 5		✓	✓
CO 6	✓	✓	

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